



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

09-06-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A	C,D	A,C	A,D	A,B,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.60 to 1.80	0.75 to 0.78	3.73 to 3.75	20.00	1.60	0.40
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	2	1	7	1	5

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C,D	A,B,C,D	C,D	B,C	B,D	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.14	3.47 to 3.48	20.00	37.50	8.00	3.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	7	8	7	1	6

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B	A,C	A,B,C,D	A,C,D	A,B,C,D	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	16.00	5.00	21.00	2.41 to 2.42	1.00	4.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	4	8	3	8	3	7

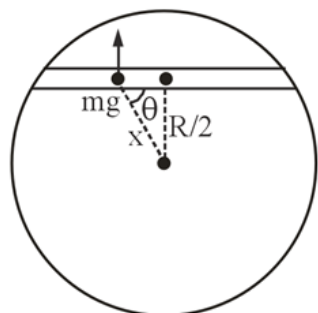
HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (A)

Net force towards centre of earth = $mg = \frac{mgx}{R}$



Normal force $N = mg \sin \theta$

Thus pressing force $N = \frac{mgx}{R} \cdot \frac{R}{2x}$

$N = \text{constant and independent of } x.$

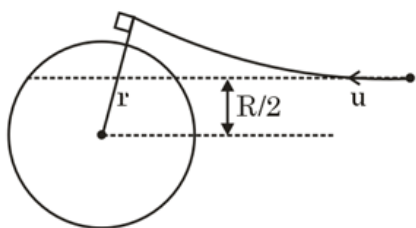
tangential force $F = ma = mg \cos \theta$

$$Q = g \cos \theta = \frac{gx}{R} \sqrt{\frac{\frac{R^2}{4} - x^2}{x}}$$

$$a = \frac{gx}{R} \sqrt{R^2 - 4x^2}$$

Curve is parabolic and at $x = \frac{R}{2}$, $a = 0$.

2. Ans (C,D)



Angular momentum conservation

$$mvr = mu \frac{R}{2}; v = \frac{uR}{2r}$$

Energy conservation.

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + \frac{kq^2}{r}$$

$$\frac{1}{2}m \left(u^2 - \frac{u^2 R^2}{4r^2} \right) = \frac{kq^2}{r}$$

$$\frac{1}{2}m \left(\frac{8kq^2}{3mR} \right) \left(1 - \frac{R^2}{4r^2} \right) = \frac{kq^2}{r}$$

$$1 - \frac{R^2}{4r^2} = \frac{6R}{8r}$$

$$8 - 2x^2 = 6x$$

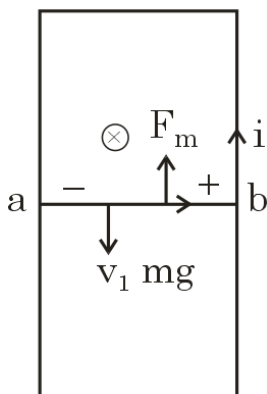
$$2x^2 + 6x - 8 = 0$$

$$2x^2 + 8x - 2x - 8 = 0$$

$$2x(x+4) - 2(x+4) = 0$$

$$x = 1 \therefore r = R$$

3. Ans (A,C)



At any instant "t"

$$i = \left(\frac{Bv\ell}{r} \right) \text{ a to b}$$

$$mg - F_m = ma$$

$$a = g - \frac{iB\ell}{m}; a = g - \left(\frac{B^2\ell^2}{mr} \right) v$$

$$\frac{dv}{dt} = g - Cv; C = \frac{B^2\ell^2}{mr}$$

$$\int_0^v \frac{dv}{g - Cv} = \int_0^t dt; v_m = \frac{g}{C} = \frac{mgr}{B^2\ell^2}$$

$$-\frac{1}{C} \ln \left[\frac{g - Cv}{g} \right] = t$$

$$1 - \frac{Cv}{g} = e^{-Ct}; v = \frac{g}{C} [1 - e^{-Ct}]$$

$$v = v_m \left[1 - e^{-\frac{gt}{v_m}} \right]; v = v_m \left[1 - e^{-\frac{gt}{v_m}} \right]$$

$$\frac{ds}{dt} = v_m \left[1 - e^{-\frac{gt}{v_m}} \right];$$

$$S = v_m \left[t - \frac{v_m}{g} e^{-\frac{gt}{v_m}} \right]$$

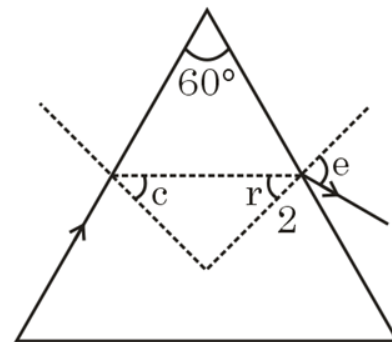
4. Ans (A,D)

$$\delta_{\min} = 2 \times 60^\circ - A \Rightarrow A = 60^\circ$$

$$r = \frac{A}{2} = 30^\circ$$

$$\sin 60^\circ = \mu \sin 30^\circ \Rightarrow \mu = \sqrt{3}$$

For $\delta_{\max}$:



$$r_2 = 60 - C$$

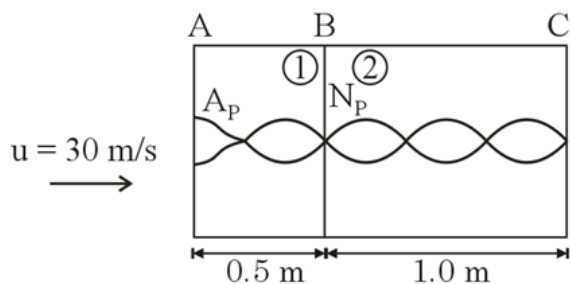
$$\sqrt{3} \sin(60 - C) = \sin e$$

$$\delta_{\max} = (90 + e) - 60^\circ$$

$$= 30^\circ + e$$

$$= 30^\circ + \sin^{-1} \left(\sqrt{\frac{3}{2}} - \frac{1}{2} \right)$$

6. Ans (B,C)



$$V_{\text{sound}} = 330 \text{ m/sec}$$

$$(i) \frac{\lambda}{4} = 0.5 \Rightarrow \lambda = 2\text{m} \rightarrow v = f\lambda \Rightarrow 1100 = f \times 2,$$

$$f = 550 \text{ Hz}$$

$$\rightarrow f' = f \left[\frac{v}{v-u} \right] = f \left[\frac{330}{300} \right] = \left(\frac{11f}{10} \right)$$

$$\rightarrow n_1 \lambda + \frac{\lambda}{2} = 0.5 \rightarrow \frac{(n_1 + \frac{1}{2})}{n_2} = \frac{1}{2}$$

$$\rightarrow n_1 = 1, n_2 = 3$$

$$n_2 \lambda = 1$$

$$\rightarrow 2\lambda + \frac{\lambda}{4} = \frac{3}{2}\text{m} \rightarrow \frac{9\lambda}{4} = \frac{3}{2}\text{m}$$

$$\lambda = \frac{2}{3}\text{m} = \frac{v}{f}$$

$$f_1 = \frac{1100}{2} \times 3; f_2 = \frac{300}{2} \times 3$$

$$= 1650 \text{ Hz} \quad = 450 \text{ Hz}$$

$$\rightarrow 11f/10 = 1650 \text{ Hz}$$

$$f = 1500 \text{ Hz}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (1.60 to 1.80)

Fundamental frequency of organ pipe

$$= \frac{v}{2\ell} = 292 \text{ Hz} = F_0$$

Fundamental frequency of wire

$$= \frac{1}{2\ell} \sqrt{\frac{T}{\rho A}} = \frac{1}{2\ell} \sqrt{\frac{\text{Stress}}{\rho}} = F_s$$

Now stress = $y \times$ strains

$$\text{Beat} = F_0 - F_s \text{ OR } F_s - F_0$$

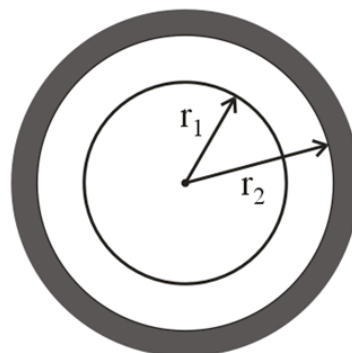
But on decreasing strain, beats also decreases

$$\text{therefore} \Rightarrow \text{Beat} = F_s - F_0 = 8$$

$$\therefore F_s = 8 + F_0 = 300 \text{ Hz} = \frac{1}{2\ell} \sqrt{\frac{y \times \text{strain}}{\rho}}$$

$$\therefore y = \frac{(300)^2 \times (2\ell)^2 \times \rho}{\text{strain}} = \frac{9 \times 4 \times 10 \times 98 \times 98 \times 10^{-4} \times 10^4}{\frac{0.02}{0.98}} = 1.694$$

2. Ans (0.75 to 0.78)



$$F = -\eta 2\pi \ell \frac{dv}{dr}$$

$$\frac{F}{2\pi \eta \ell} \int_{R_2}^{R_1} \frac{dr}{r} = - \int_0^{\omega R_1} dv$$

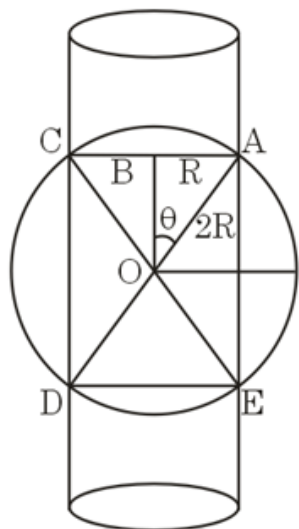
$$\frac{F}{2\pi \eta \ell} \ln \frac{R_1}{R_2} = -\omega R_1$$

$$F = \frac{2\pi \eta \ell \omega R_1}{\ln \left(\frac{R_2}{R_1} \right)}$$

$$\tau = F R_1 = \frac{2\pi \eta \ell \omega R_1^2}{\ln \left(\frac{R_2}{R_1} \right)}$$

$$\therefore \eta \approx 0.76$$

3. Ans (3.73 to 3.75)



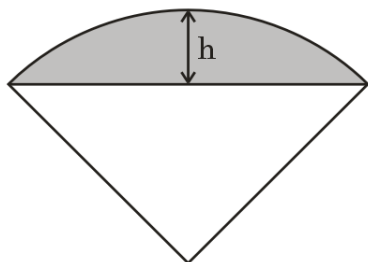
$$\theta = 35^\circ$$

$$OB = 2R \cos \theta = R\sqrt{3}$$

enclosed volume = volume of cylinder ACDE

volume of two spherical cap

vol of spherical caps



$$V_{\text{cap}} = \pi h^2 \left(R - \frac{h}{3} \right)$$

$$\text{here } h = 2R - 2R \cos \theta$$

$$= 2R - R\sqrt{3}$$

$$\text{So } V_{\text{cap}} = \pi(2R - R\sqrt{3})^2 \left(2R - \frac{2R - R\sqrt{3}}{3} \right)$$

$$= \left(\frac{16}{3} - 3\sqrt{3} \right) \pi R^3$$

$$\text{Total volume} = (\pi R^2) 2R\sqrt{3} + 2 \times \left(\frac{16}{3} - 3\sqrt{3} \right) \pi R^2$$

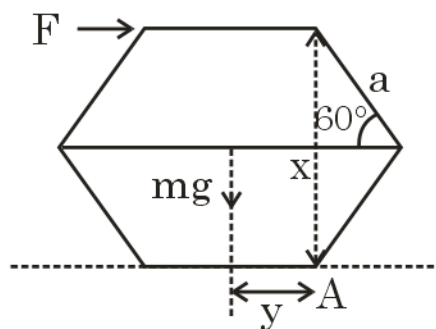
$$= 2\pi R^3 \left(\sqrt{3} + \frac{16}{3} - 3\sqrt{3} \right)$$

$$= 2\pi R^3 \left(\frac{16 - 6\sqrt{3}}{3} \right)$$

$$d = \frac{\rho V}{\epsilon_0} = V = \frac{2}{3} \pi R^3 [16 - 6\sqrt{3}]$$

$$= \frac{2}{3} \pi R \frac{1}{\pi} [16 - 6\sqrt{3}]$$

4. Ans (20.00)



$$x = 2a \sin 60^\circ = \sqrt{3}a$$

$$y = a - a \cos 60^\circ = \frac{a}{2}$$

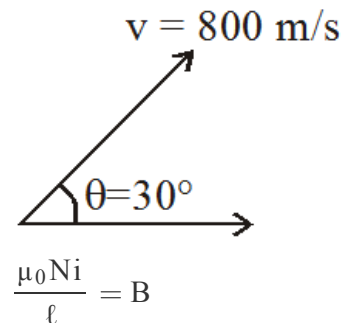
For toppling

$$Fx \geq mgy$$

$$F \geq \frac{mgy}{x} = \frac{mg \frac{a}{2}}{\sqrt{3}a} = \frac{mg}{2\sqrt{3}}$$

$$F \geq 20N$$

5. Ans (1.60)



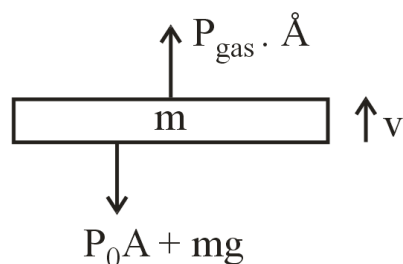
$$X = \text{Pitch} = (v \cos 30^\circ) \frac{2\pi m}{qB} = \frac{v\sqrt{3}\pi m \ell}{\mu_0 Ni q}$$

$$\text{no. of turns} = \frac{\ell}{n} = \frac{\mu_0 Ni}{\sqrt{3}v\pi m} \left(\frac{q}{m} \right)$$

$$= \frac{4\pi \times 10^{-7} \times 8000 \times 4 \times \sqrt{3} \times 10^{11}}{\sqrt{3} \times 800 \times \pi}$$

$$= 16 \times 10^5 = 1.6$$

6. Ans (0.40)



$$P_{\text{gas}} = P_0 + \frac{mg}{A}$$

$$Q = \frac{F}{2} n R \Delta T + P \Delta v$$

$$\frac{dQ}{dt} = \left(\frac{F}{2} + 1 \right) P \Delta v = \left(\frac{F}{2} + 1 \right) P A \cdot \frac{dx}{dt}$$

$$v = \frac{q}{\left(\frac{F}{2} + 1 \right) P A} = \frac{q}{\frac{5}{2} \times [P_0 A + mg]}$$

$$v = \frac{2}{5} \frac{q}{(mg + P_0 A)} \Rightarrow \frac{x}{y} = 0.4$$

PART-1 : PHYSICS

SECTION-III

1. Ans (3)

$$\text{Angular width} = \frac{\lambda}{d} = 10^{-3} \text{ (given)}$$

∴ Number of fringes within 0.12° will be

$$n = \frac{0.12 \times 2\pi}{360 \times 10^{-3}} \cong [2.09]$$

∴ The number of bright spots will be three.

2. Ans (2)

Half life = 693.1 days

$$\frac{0.693}{\lambda} = 693 \times 24 \times 3600$$

$$\Rightarrow \lambda = \frac{1}{1000 \times 24 \times 3600} \text{ s}^{-1}$$

$$= \frac{1}{24 \times 36} \times 10^{-5}$$

$$E = 8.4 \times 10^{-14} \text{ J}$$

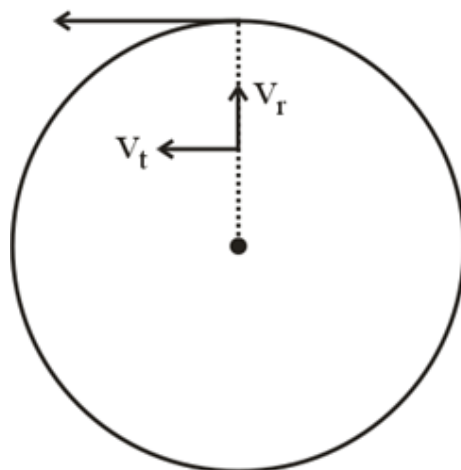
$$\Rightarrow \frac{12.6}{100} \times \lambda N \times E = 441 \times 10^3$$

$$N = \frac{441 \times 10^5 \times 24 \times 36}{12.6 \times 8.4 \times 10^{-19}} = 3600 \times 10^{24}$$

$$\Rightarrow \frac{N}{N_A} = \frac{36}{6.023} \times \frac{10^{26}}{10^{23}} \approx 6 \times 10^3$$

$$= \frac{n}{m} = \frac{6}{3} = 2$$

3. Ans (1)



$$v_t = r \left(\frac{u}{R} \right)$$

$$v_t^2 + v_r^2 = u^2$$

$$\left(\frac{r}{7} \right)^2 + v_r^2 = 16$$

$$\frac{dr}{dt} = \sqrt{16 - \frac{r^2}{7^2}}$$

$$7 \int_0^R \frac{dr}{\sqrt{(28)^2 - r^2}} = \int dt$$

$$7 \left(\sin^{-1} \left(\frac{R}{28} \right) \right) = t$$

$$t = \frac{7\pi}{2} = 11 \text{ sec, Ans} = 1.00$$

4. Ans (7)

$$\text{S} \xrightarrow{1\text{m}} 1 \text{ cm}^2$$

$$N_0 = 3 \times 10^{16}; A = 50000 \text{ counts/sec.}$$

$$N = N_0 e^{-\lambda t}; \lambda N = \frac{-dN}{dt}$$

$$\frac{1 \times 10^{-4}}{4\pi(1)^2} \times \lambda(3 \times 10^{16}) = 50000$$

$$\frac{3\lambda \times 10^{12}}{4\pi} = 5 \times 10^4$$

$$\lambda = \frac{20\pi}{3} \times 10^{-8} \approx 21 \times 10^{-8}$$

$$= 3 \times 7 \times 10^{-(7+1)}$$

$$n = 7$$

5. Ans (1)

$$q' = \frac{qd}{2\pi R} \rightarrow y = \frac{-\frac{qd}{2\pi R} \cdot R}{q} = \left(\frac{-d}{2\pi} \right)$$

$$\tau = q \cdot \frac{d}{2\pi} \times E = \text{initial torque}$$

$$\tau_{\text{gen}} = qE \frac{d}{2\pi} \cos \theta$$

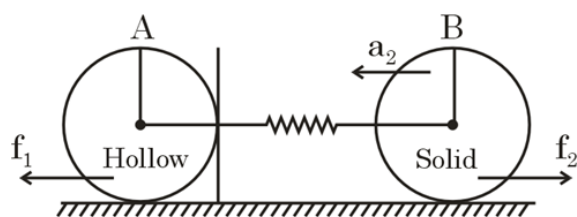
$$W = q \frac{Ed}{2\pi} \int_0^{\frac{\pi}{2}} \cos \theta d\theta = \frac{1}{2} I_{\text{com}} \omega^2$$

$$q \frac{Ed}{2\pi} \cdot (1) = \frac{1}{2} mR^2 \cdot \omega^2$$

$$\omega = \sqrt{\frac{qEd}{\pi mR^2}} = \frac{1}{R} \cdot \sqrt{\frac{qEd}{\pi m}}$$

$$x = 1, y = 1 \Rightarrow \frac{y}{x} = 1$$

6. Ans (5)



Let x_1 and x_2 be displacement from equilibrium position for hollow

$$k(x_1 + x_2)r = \frac{5}{3}mr^2 - \frac{a_1}{r} \dots (i)$$

Angular momentum around (A) of system

$$\frac{5}{3}mv_1r = \frac{7}{5}mv_2r$$

$$25v_1 = 21v_2 \dots (ii)$$

$$25x_1 = 21x_2 \dots (iii)$$

using (i) and (iii)

$$k \left(x_1 + \frac{25}{21}x_1 \right) = \frac{5}{3}ma_1 \Rightarrow a_1 = \frac{46}{35} \frac{k}{m} x_1$$

$$f = \frac{1}{2\pi} \sqrt{\frac{46}{35}} k$$

$$A_1 + A_2 = x_0 \dots (iv)$$

$$A_2 = \frac{25}{21}A_1 \dots (v)$$

by (iv) and (v)

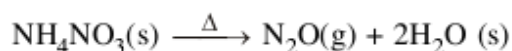
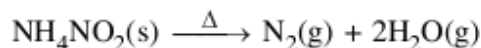
$$A_1 = \frac{21}{46}x_0; A_2 = \frac{25}{46}x_0$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (B,C,D)

N_2O is a linear molecule. Its is a laughing gas



3. Ans (C,D)

$$t_{1/2} \propto [A]^{1-n}$$

$$1 - n = -2 \Rightarrow n = 3$$



$$t = 0 \quad 2 \quad 1 \quad -$$

$$t = t \quad 2-2x \quad 1-x \quad x$$

$$-\frac{d[B]}{dt} = K(2-2x)^3(1-x)^m$$

$$= K \cdot 2^3(1-x)^{m+3}$$

$$m + 3 = 2$$

$$m = -1$$

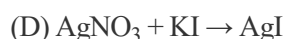
4. Ans (B,C)

(A) Adsorption is enthalpy driven process ($\Delta H < 0$)

(B) Promoter increases catalytic activity

(C) Muddy water is -ve charged and coagulation

power of $Al^{3+} > Na^+$



AgI/Ag^+ sol is formed.

which migrate towards cathode in electric field

PART-2 : CHEMISTRY

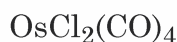
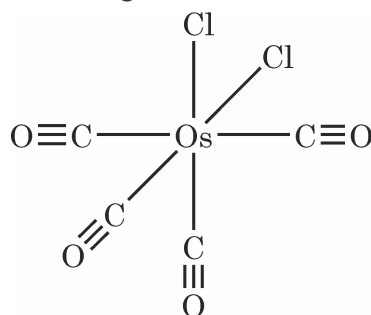
SECTION-II

1. **Ans (1.14)**

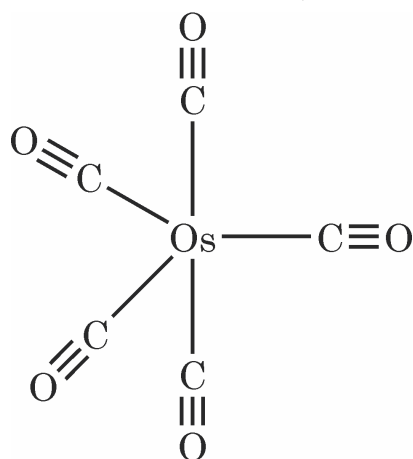
$x = 1 \Rightarrow$ No possibility of geometrical isomerism in 1st complex.

$x = 2 \Rightarrow$ Possible with octahedral & TBP geometries.

$x = 3 \Rightarrow$ Geometry of 1st complex will not have 90° & 180° angles.



$\Rightarrow$ cis-isomer octahedral, P = 8 atoms



$\Rightarrow$ TBP geometry, Q = 7 atoms in equatorial plane.

2. **Ans (3.47 to 3.48)**

PCl_5 and PBr_5 both exist in ionic form in solid state, both give H_3PO_4 and HX on hydrolysis.

Bromine is liberated from NaBr on reaction with conc H_2SO_4 (this is not possible with NaCl).

Hence, $a = 5$, $b = 3$, $c = 4$ and atomic weight of Bromine is 80 amu. Hence 80/23 value is 3.48

3. **Ans (20.00)**

$$dH = dU + d(PV)$$

$$dH = dU + V.dP$$

$$P = \frac{nRT}{V - nb} - \frac{an^2}{V^2}$$

$$dP = \frac{nRdT}{(V - nb)^2}$$

$$dH = dU + V \times \frac{nR dT}{(V - nb)^2}$$

$$\frac{310}{9} \times 10^3 = 30 \times 10^3 + \frac{n \times 2 \times 20 \times 100}{(20 - 0.1n)}$$

$$\frac{310}{9} - 30 = \frac{4n}{(20 - 0.1n)}$$

$$\frac{40}{9} = \frac{4n}{(20 - 0.1n)}$$

$$200 - n = 9n$$

$$n = 20$$

4. **Ans (37.50)**

$$P = \frac{h}{\lambda}$$

$$\Delta P = \frac{h}{\lambda^2} \Delta \lambda$$

$$\Delta P \cdot \Delta x \geq \frac{h}{4\pi}$$

$$\Delta \lambda \geq \frac{\lambda^2}{4\pi} \left(\frac{1}{\Delta x} \right)$$

($y = mx$ form)

$$\text{slope} = \tan \theta = \frac{\lambda^2}{4\pi} = \frac{1}{\pi}$$

$$\lambda = 2\text{\AA}$$

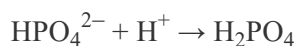
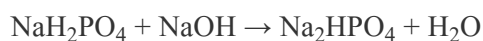
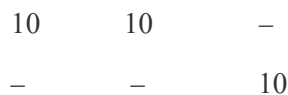
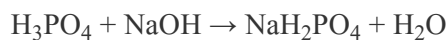
$$\lambda(\text{\AA}) = \sqrt{\frac{150}{V}} = 2$$

$$V = \frac{150}{4} = 37.5 \text{ V}$$

PART-2 : CHEMISTRY

SECTION-III

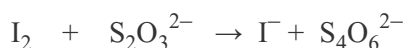
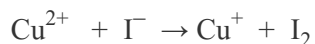
3. Ans (8)



$$\text{pH} = \text{p}K_{a2} + \log \frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]}$$

$$= \text{p}K_{a2} = 8$$

4. Ans (7)



$$= 0.1 \times 12.12 = 1.212$$

$$= \text{meq. of } \text{I}_2$$

$$= \text{meq. of } \text{Cu}^{2+}$$

$$\text{moles of } \text{Cu}^{2+} = 1.212 \times 10^{-3}$$

$$\text{weight of } \text{Cu}^{2+} = 1.212 \times 63.5 \times 10^{-3}$$

$$\% \text{ Cu} = \frac{1.212 \times 63.5 \times 10^{-3}}{1.1} \times 100 = 7$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (A,B)

$$\int_0^1 g(x) dx = \frac{11}{6}$$

$$\Rightarrow \frac{a}{3} + \frac{b}{2} + c = \frac{11}{6}$$

$$\Rightarrow 2a + 3b + 6c = 11$$

$$\therefore a = b = c = 1 \quad \{ \because a, b, c \in \mathbb{N} \}$$

$$\therefore \frac{f(x)f'(x)}{\sqrt{1-(f(x))^4}} - x \geq 0$$

$$\Rightarrow h(x) = \sin^{-1}(f(x)^2) - x^2 \text{ is non-decreasing}$$

function

$$\therefore \lim_{x \rightarrow x_1^+} h(x) \leq \lim_{x \rightarrow x_2^-} h(x)$$

$$\Rightarrow \frac{\pi}{2} - x_1^2 \leq \frac{\pi}{6} - x_2^2$$

$$\Rightarrow x_1^2 - x_2^2 \geq \frac{\pi}{3}$$

$$\Rightarrow [x_1^2 - x_2^2] \geq 1$$

$$\text{i.e., } [x_1^2 - x_2^2] = 1$$

2. Ans (A,C)

Equation of normal chord at $P(2t_1^2, 4t_1)$ & $R(2t_2^2, 4t_2)$

$$\text{and } y + xt_2 - 4t_2 - 2t_2^3 = 0.$$

Equation of any two degree curve passing through P,

Q, R, S is

$$(y + xt_1 - 4t_1 - 2t_1^3)(y + xt_2 - 4t_2 - 2t_2^3) + \lambda(y^2 - 8x)$$

$$= 0$$

$\therefore$ P, Q, R, S are con cyclic

$$\therefore 1 + \lambda = t_1 t_2 \text{ \& } t_1 + t_2 = 0$$

$\therefore$ Centroid of ΔAPR lies on x-axis & slope of

$$\text{PR} = \frac{4(t_1 - t_2)}{2(t_1^2 - t_2^2)} = \frac{2}{t_1 + t_2}$$

PR is parallel to y-axis

3. Ans (A,B,C,D)

(A) In skew symmetric matrix $a_{ij} = -a_{ji}$ and all element of diagonal are zero

$$\text{So required probability} = \frac{3 \times 3 \times 3}{4^9} = \frac{27}{4^9}$$

(B) For symmetric matrix $a_{ij} = a_{ji}$

So, required probability =

$$\frac{(4)(4)(4)(4)(4)(4)(1.1.1)}{4^9} = \frac{1}{64}$$

$$(C) \text{ Probability } \frac{4!}{4^4} = \frac{3}{32}$$

(D) Probability

$$= \frac{4! \left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right)}{4!} = \frac{9}{24} = \frac{3}{8}$$

4. Ans (A,C,D)

$$P_k = e^{i\theta_k} \quad \theta_k = \frac{2k\pi}{n}$$

$$A_k = x_k + iy_k = a \cos \theta_k + i b \sin \theta_k$$

$$\frac{x_k}{a} = \cos \theta_k, \quad \frac{y_k}{b} = \sin \theta_k$$

$$\frac{x_k^2}{a^2} + \frac{y_k^2}{b^2} = 1 \Rightarrow A_k \text{ lies on ellipse}$$

$$\text{Now } \sum_{k=1}^n (A_k S)^2 = \sum_{k=1}^n (a - ex_k)^2$$

$$= na - e \sum_{k=1}^n x_k = na$$

$$= na^2 + a^2 e^2 \sum_{k=1}^n \cos^2 \theta_k$$

$$= na^2 + \frac{1}{2} a^2 e^2 \sum_{k=1}^n \cos^2 \theta_k$$

$$\left(\sum_{k=1}^n \cos^2 \theta_k = 0 \text{ sum of the root of } z^{n/2} - 1 = 0 \right)$$

$$= na^2 + \frac{na^2}{2} \left(1 - \frac{b^2}{a^2} \right) = \frac{n}{2} (3a^2 - b^2)$$

5. Ans (A,B,C,D)

$$(A) \int_1^2 \frac{\sin x}{x^2} < \int_1^2 \frac{x}{x^2} dx = \ln 2$$

$$(B) \int_1^2 \frac{\sin x}{x^2} dx > \int_1^2 \frac{1}{x} - \frac{x}{6} dx = \ln 2 - \frac{1}{4}$$

$$(C) \int_1^2 \frac{\sin x}{x^2} dx > \int_1^2 \frac{1}{\sqrt{2} x^2} dx = \frac{1}{2\sqrt{2}}$$

$$(D) \int_1^2 \frac{\sin x}{x^2} dx > \int_1^{\frac{2\pi}{x}} \frac{2x}{x^2} dx + \int_{\frac{2\pi}{x}}^2 \frac{-\frac{2}{x}(x-\pi)}{x^2} dx$$

$$= \frac{2}{\pi} \ln \pi - \left[\frac{2}{\pi} \ln \left(\frac{2}{\pi} \right) + 2 \left[\frac{1}{2} - \frac{1}{\pi} \right] \right]$$

$$= \frac{2}{\pi} \ln \left(\frac{\pi^2}{2} \right) - \left(\frac{\pi-2}{\pi} \right)$$

6. Ans (A,C,D)

$$\text{Volume} = \frac{2\sqrt{2}}{3}$$

$$\Rightarrow \frac{1}{3} \cdot \frac{1}{2} \left\| \begin{matrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 1 \\ 2 & -1 & -1 \end{matrix} \right\| \times h = \frac{2\sqrt{2}}{3}$$

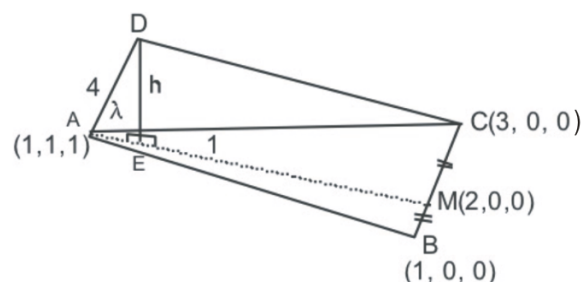
$$\Rightarrow h |\hat{j} - \hat{k}| = 2\sqrt{2} \Rightarrow h = 2$$

for E, Let AE : EM = $\lambda : 1$

$$\Rightarrow E \equiv \left(\frac{2\lambda+1}{\lambda+1}, \frac{1}{\lambda+1}, \frac{1}{\lambda+1} \right) \&$$

$$(AE)^2 + (ED)^2 = (AD)^2$$

$$\Rightarrow \lambda = -2 \text{ or } \lambda = -2/3$$



PART-3 : MATHEMATICS

SECTION-II

1. Ans (16.00)

We have

$$f(x) = e + (1-x)\ln\left(\frac{x}{e}\right) + \int_1^x f(t) dt \quad \dots(i)$$

Differentiable both sides with respect to x, we get

$$f'(x) = (1-x)\frac{1}{x} + \ln\left(\frac{x}{e}\right)(-1) + f(x)$$

$$\Rightarrow f'(x) - f(x) = \left(\frac{1}{x} - 1 - \ln\left(\frac{x}{e}\right)\right)$$

$\therefore$ Multiplying both sides with e^{-x} , we get

$$e^{-x}f'(x) - e^{-x}f(x) = e^{-x}\left(\frac{1}{x} - 1 - \ln\left(\frac{x}{e}\right)\right)$$

$$\Rightarrow \frac{d}{dx}(e^{-x} \cdot f(x)) = e^{-x}\left(\frac{1}{x} - 1 - \ln\left(\frac{x}{e}\right)\right)$$

$\therefore$ On integrating both sides with respect to x,

we get

$$e^{-x}f(x) = e^{-x} - \int e^{-x}\left(\ln\left(\frac{x}{e}\right) - \frac{1}{x}\right) dx$$

$$= e^{-x} + e^{-x} \cdot \ln\left(\frac{x}{e}\right) + \lambda$$

$$\Rightarrow f(x) = 1 + \ln\left(\frac{x}{e}\right) + \lambda e^x \quad \dots(ii)$$

Now, putting $x = 1$ in equation (i),

$$\text{we get } f(1) = e \quad \dots(iii)$$

$\therefore$ Using (iii) and (ii), we get $\lambda = 1$

$$\text{Thus, } f(x) = 1 + e^x + \ln\left(\frac{x}{e}\right) \text{ or } f(x) = e^x + \ln x$$

$$\text{Hence, } g(x) = x \ln x; \text{ now } A = \int_0^1 x \ln x dx = \frac{1}{4}$$

$$A^{-2} = \left(\frac{1}{4}\right)^{-2} = 16$$

2. Ans (5.00)

$$\vec{p} \cdot \vec{q} = 3^{2 \sin x + \tan x - 1} + 3^{\sin x + 2 \tan x - 1} - 3^{\sin x + \tan x}$$

$$3\vec{p} \cdot \vec{q} = 3^{2 \sin x + \tan x} + 3^{\sin x + 2 \tan x} - 3 \cdot 3^{\sin x + \tan x}$$

$$3\vec{p} \cdot \vec{q} + 1 = \left(3^{\left(\frac{2 \sin x + \tan x}{3}\right)}\right)^3 + \left(3^{\frac{\sin x + 2 \tan x}{3}}\right)^3 + 1^3$$

$$- 3 \cdot \left(3^{\frac{2 \sin x + \tan x}{3}}\right) + \left(3^{\frac{\sin x + 2 \tan x}{3}}\right) \cdot 1$$

If a, b, c are positive then

$$a^3 + b^3 + c^3 - 3abc \geq 0$$

and equality holds when $a = b = c$

$$\therefore \sin x = \tan x = 0 \Rightarrow x = x\pi$$

$$S = 0 + \pi + 2\pi + \dots + 50\pi = \frac{\pi(50 \times 51)}{(2)}$$

$$\therefore \frac{S}{255\pi} = 5$$

3. Ans (21.00)

$$f(x) = ([x] + [|x + \sin x|])\operatorname{sgn}(x^2 - 1)$$

$|x + \sin x|$ will have 11 points of discontinuity

$\{x\}$ will have 11 points of discontinuity of

which $x = 0$

is common in both and $f(x)$ is discontinuous at

$$x = \pm 1$$

4. Ans (2.41 to 2.42)

$$A_1 = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_I + \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}_B \right)^n$$

$$A = (I + B)^n = {}^nC_0 I + {}^nC_1 B + {}^nC_2 B^2 + {}^nC_3 B^3 + \dots + {}^nC_n B^n$$

Since,

$$B^2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$$

$$B^{2k} = 2^k I \text{ and } B^{2k+1} = B^{2k} \cdot B = 2^k \cdot B$$

$$\Rightarrow A = \underbrace{\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n}_{x} = \underbrace{({}^nC_0 + {}^nC_2 2 + {}^nC_4 2^2 + \dots)}_x I + \underbrace{[{}^nC_1 + {}^nC_3 2 + {}^nC_5 2^2 + \dots]}_y B$$

$$= \begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & y \\ y & -y \end{bmatrix} = \begin{bmatrix} x+y & y \\ y & x-y \end{bmatrix}$$

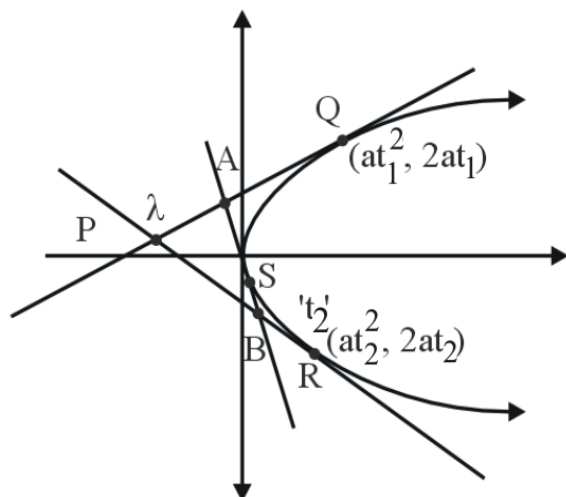
$$\frac{a_{12}}{a_{22}} = \frac{y}{x-y}$$

$$= \frac{({}^nC_1 + {}^nC_3 2 + {}^nC_5 2^2 + \dots)}{({}^nC_0 + {}^nC_2 2 + {}^nC_4 2^2 + \dots) - ({}^nC_1 + {}^nC_3 + \dots)}$$

$$\frac{\frac{1}{\sqrt{2}}((1+\sqrt{2})^n - (1-\sqrt{2})^n)}{2} = 1 + \sqrt{2}$$

$$\lim_{n \rightarrow \infty} \frac{2}{(\sqrt{2}-1)(1+\sqrt{2})^n - (\sqrt{2}+1)(1-\sqrt{2})^n} = 1 + \sqrt{2}$$

5. Ans (1.00)



$$P(at_1 t_2, a(t_1 + t_2))$$

$$S(at_3^2, 2at_3)$$

$$A(at_1 t_3, a(t_1 + t_3))$$

$$B(at_2 t_3, a(t_2 + t_3))$$

$$\text{Let } PA : PQ = \lambda : 1$$

Using section formula

$$\lambda = \frac{t_3 - t_2}{t_1 - t_2}$$

$$\Rightarrow \frac{PA}{PQ} = \frac{t_3 - t_2}{t_1 - t_2}$$

$$\text{Similarly } \frac{PB}{PR} = \frac{t_1 - t_3}{t_1 - t_2}$$

$$\Rightarrow \frac{PA}{PQ} + \frac{PB}{PR} = 1$$

6. Ans (4.00)

Put $y = x$

$$\Rightarrow f(x^2) = 2xf(x)$$

$$\therefore \int_0^x f(x) dx = \frac{f(x^2) - x^2}{4}$$

$$\Rightarrow \int_0^x f(x) dx = \frac{2xf(x) - x^2}{4}$$

$\therefore$ Differentiable w.r.t. x

$$\Rightarrow \frac{dy}{dx} + \left(-\frac{1}{x}\right)y = 1$$

$$\text{As } f(1) = 0$$

$$\Rightarrow f(x) = x \ln x$$

$$\therefore s = - \int_0^1 x \cdot \ln x dx = \frac{1}{4}$$

$$\Rightarrow \log_{\sqrt{6}} \left(\frac{9}{5}\right) = \log_{\sqrt{6}} (36) = 4$$

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (4)**

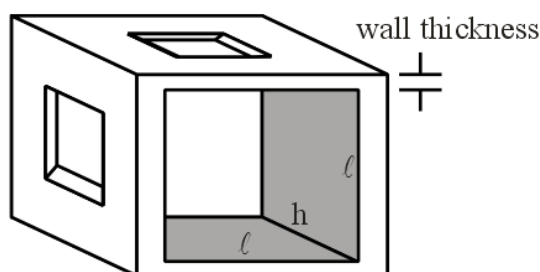
$$4\sin^4 x + 4\sin^3 x + 4\sin^2 x = \sin^2 x + \sin x + 1$$

$$\Rightarrow 4\sin^2 x(\sin^2 x + \sin x + 1) = \sin^2 x + \sin x + 1$$

$$\Rightarrow \sin x = \pm \frac{1}{2}$$

2. **Ans (8)**

Let box dimension be $\ell \times \ell \times h$.



$$\text{So, } \ell^2 h = 8^3$$

$$V = (\ell + 4)^2(h + 4) - \ell^2 h - (2\ell^2 + 4\ell h) \times \frac{1}{2} \times 2$$

$$= 2 \left(\ell^2 + 16\ell + \frac{2 \times 8^3}{\ell} + \frac{8^4}{\ell^2} + 32 \right)$$

$$= 2 \left(\ell^2 + 8\ell + 8\ell + \frac{8^3}{\ell} + \frac{8^3}{\ell} + \frac{8^4}{\ell^2} + 32 \right) \geq 2 \times (6 \times 8^2 + 32)$$

($\because$ A.M. $\geq$ G.M.)

Further, equality holds when $\ell = 8$

i.e. $h = 8$

3. **Ans (3)**

$$T_r = \frac{(2r+1)(2r^2+2r-1)}{(r^2-1)^2(r^2+2r)^2} = \frac{(2r+1)(2r^2+2r-1)}{(r^2-1)^2((r+1)^2-1)^2}$$

$$\therefore T_r = \left[\frac{1}{(r^2-1)^2} - \frac{1}{((r+1)^2-1)^2} \right]$$

$$\therefore K = \lim_{n \rightarrow \infty} \sum \left(\left(\frac{1}{(2^2-1)^2} - \frac{1}{(3^2-1)^2} \right) + \left(\frac{1}{(3^2-1)^2} - \frac{1}{(4^2-1)^2} \right) + \dots + \left(\frac{1}{(n^2-1)^2} - \frac{1}{((n+1)^2-1)^2} \right) \right)$$

$$\therefore K = \frac{1}{9}$$

4. **Ans (8)**

$$Z^{2n+1} = e^{2\pi i} = 1$$

$$A + B = z + z^2 + \dots + z^{2n-1} + z^{2n}$$

$$= 1 + z + z^2 + \dots + z^{2n} - 1 = 0 - 1 = -1$$

$$AB = (z + z^3 + \dots + z^{2n-1})(z)(z + z^3 + \dots + z^{2n-1})$$

$$= z(z + z^3 + z^5 + \dots + z^{2n-1})^2$$

$$= z \left[\frac{z(1-z^{2n})}{1-z^2} \right]^2 = z \left(\frac{z-1}{1-z^2} \right)^2 = \frac{z}{(1+z)^2}$$

$$\text{let } z = e^{2\theta i}, \theta = \frac{\pi}{2n+1}$$

$$AB = \frac{e^{2\theta i}}{(1 + e^{2\theta i})^2} = \frac{1}{(e^{-\theta i} + e^{\theta i})^2}$$

$$= \frac{1}{4} \sec^2 \left(\frac{\pi}{2n+1} \right)$$

hence quadratic equation is

$$x^2 - x(A+B) + AB = 0$$

$$\Rightarrow 4x^2 + 4x + \sec^2 \left(\frac{\pi}{2n+1} \right) = 0$$

5. Ans (3)

$$f'g - 2g^2f^2 = 1 \quad \dots\dots(i)$$

$$g'f - 2f^2g^2 = 3 \quad \dots\dots(ii)$$

adding the two equations

$$\text{add } f'g + g'f - 4f^2g^2 = 4$$

$$\frac{d}{dx}(f(x).g(x)) = 4(1 + f^2g^2)$$

$$\int \frac{d(f(x)g(x))}{1 + (f(x)g(x))^2} = 4 \int dx$$

$$\tan^{-1}(f(x).g(x)) = 4x + c$$

$$(\text{here put } x = 0) \Rightarrow \tan^{-1} 1 = c \Rightarrow c = \frac{\pi}{4}$$

$$f(x)g(x) = \tan\left(4x + \frac{\pi}{4}\right)$$

$$A = \int_{-\pi/48}^{\pi/48} \tan\left(4x + \frac{\pi}{4}\right) dx$$

$$= \int_{-\pi/48}^{\pi/48} \frac{1 + \tan 4x}{1 - \tan 4x} dx$$

$$A = \int_0^{\pi/48} \left(\frac{1 + \tan 4x}{1 - \tan 4x} + \frac{1 - \tan 4x}{1 + \tan 4x} \right) dx$$

$$= 2 \int_0^{\pi/48} \frac{dx}{\cos 8x}$$

$$\Rightarrow \text{put } 8x = t, \frac{dt}{dx} = 8$$

$$A = \frac{2}{8} \int_0^{\pi/6} \frac{dt}{\cos t} \Rightarrow \frac{1}{4} (\log(\sec t + \tan t)) \Big|_0^{\pi/6}$$

$$A = \frac{1}{4} [\log(\sqrt{3})]$$

$$\Rightarrow A = \log 3^{1/8} \Rightarrow 8A = \log 3 \Rightarrow e^{8A} = 3$$

6. Ans (7)

$$\triangle CGP \sim \triangle CNB \Rightarrow CP : PB = 2 : 1$$

$$\text{By section formula } \frac{b+1+2a}{3} = 2$$

$$\Rightarrow 2a + b + 1 = 6 \quad \dots\dots(1)$$

$$\text{Similarly } AQ : QB = 2 : 1$$

$$\Rightarrow k + 2b = 3 \quad \dots\dots(2)$$

From (1) and (2),

we get

$$k + 2(5 - 2a) = 3 \Rightarrow k = 4a - 7$$

$$\text{Slope of median AG} = \frac{k-1}{a-2} = \frac{4a-8}{a-2} = 4$$

$$\Rightarrow \text{equation of AG is } y - 1 = 4(x - 2) \Rightarrow y = 4x - 7$$

